

Sustainability Strategy



SDG sub-targets
9.1, 9.4

NORDEN's Executive Body for Sustainability, chaired by NORDEN's CEO, is ultimately responsible for sustainability across the Group. The Executive Body for Sustainability is cross-functionally represented and is responsible for mandating NORDEN's Sustainability Strategy, ambition level and policies as recommended by the Sustainability Department.

Launched in 2019, NORDEN's Sustainability Strategy focuses on finding new business opportunities within the agenda of the Sustainable Development Goals (SDGs) where NORDEN can contribute to a significant difference. This is an addition to NORDEN's operational foundation where we work with responsibility and sustainability. In the strategy development process, NORDEN has prioritised the SDGs and subsequently mapped them to our core business activities.

Sustainable Development Goals

Adopted by the UN in

2015

17 goals and 169 underlying targets define global sustainable development priorities and aspirations for the year

2030

For NORDEN, the SDGs are chosen as basis for our Sustainability Strategy

2019-21

SDG 9 Industry, Innovation and Infrastructure holds a strategic potential aligned with NORDEN's business strategy

2 areas of business opportunity

Decarbonisation

Shipping has a high climate impact. At NORDEN, we have come a long way in increasing fuel efficiency. With the IMO targets on CO₂ reductions, it is evident that increased fuel efficiency alone is not enough. We need alternative solutions, and under the Decarbonisation heading, we want to contribute with low-carbon transportation options.

Decarbonisation performance 2019

NORDEN has expanded its portfolio of initiatives. We are currently looking into different alternative fuel options. We have already completed 2 test voyages with 100% biofuel from Good-fuels and are now working on the last adjustments. We are also working with pioneer company Kvasir Technologies on a bio crude oil solution, and further, we are investigating a wing sails solution. We expect to engage in even more partnerships in the future.

Logistics optimisation

Effective transport and logistics can help drive the expansion of world trade. We will use NORDEN's knowledge and experience in optimising infrastructure logistics by addressing bottleneck issues with for instance congestion or inefficiencies in port handling, unpredictability in inland transportation or difficulties in transitions from inland transportation to ocean-going vessels. Under the Logistics optimisation heading, we will find collaborative ways to remove these barriers, creating shared value for customers, partners, local communities, regions and NORDEN.

Logistics optimisation performance 2019

NORDEN is currently in the process of developing 4 port optimisation projects around the world to become increasingly involved with cargo loading and unloading operations. With our port captains, we are looking at optimising operation and hope to contribute to the development of local infrastructure.

Climate & Environment



SDG sub-targets
9.1, 9.4, 12.4, 12.7, 12c,
13.2, 13.3, 14.1, 14.3, 14c

Climate & Environment covers efficiency and various emissions

Climate

Climate change derived from CO₂ emissions is a risk to biodiversity and livelihoods and poses operational and commercial challenges to NORDEN's business.

NORDEN's climate goal is aligned with IMO's carbon reduction strategy in reducing relative CO₂ emissions by 40% in 2030 compared to 2008. Further, in December 2019, the Danish government's climate partnership with the Danish maritime sector defined a climate ambition of a 2050 carbon neutrality. NORDEN applauds this ambition, and under our Sustainability Strategy, we will also work towards this goal by engaging in partnerships.

NORDEN's activity measured in transport work (tonnes cargo transported) increased by 7% in 2019 compared to 2018. This is

reflected in our CO₂ emissions, which also increased by 7% compared to 2018.

The last few years, the relative emission reductions have not increased much. The year 2019 shows we have reached 22% of our total 2030 relative emission reduction target. NORDEN has worked to optimise operational air emissions for several years, but it is clear that we need to do more. Our Advanced Analytics and Digital Solutions Department works with a strategic ambition of investigating new low-carbon fuel options as well as optimising logistics, and NORDEN wants to further add to the decarbonisation agenda as is stated in our Sustainability Strategy. We have several decarbonisation initiatives in the pipeline that will hopefully address this development in a positive direction during 2020.

NORDEN's climate target is aligned with the targets of IMO and the Danish government

40%

40% transport work emission reduction by 2030 compared to 2008

100%

100% carbon neutral in 2050

8.70

EEOI

Climate

NORDEN's ambition

- Further investigate low-carbon shipping options and logistical optimisation at ports
- Transport work emission reduction of 40% by 2030 compared to 2008
- Carbon neutral in 2050 aligned with Danish Shipping and the Danish government's climate partnership

Performance 2019

Transport work

- 470,000 million tonne-miles (tonnes cargo transported 1 mile), an increase of 7% compared to 2018

EEOI (Energy Efficiency Operational Indicator)

- 8.70, a slight increase from 8.60 in 2018.
- Compared to 2008: All operated vessels reached 22% towards the 2030 target, i.e. we have 10 years to make up for the remaining 78%. Assuming a linear reduction, NORDEN has come 45% of where we should be today

Emissions

- CO₂: 4,092,000 tonnes

Absolute CO₂ reduction

- Accounting for absolute CO₂ reduction will commence in 2020 under our decarbonisation initiative.

Looking ahead

- Focus on further fuel efficiency and low-carbon options

Relevant policies

- Safety & Environmental Protection Policy: All employees in every function execute their work under safe and healthy conditions and with proper concern for protecting the environment. NORDEN aims to eliminate possible dangers which may result in personal injury, professional illness, accidents, damage to property and damage to the environment
- Climate Policy for Danish Shipping from November 2019 describes expectations for the shipping industry, including a zero carbon industry in 2050 and the first carbon neutral vessels by 2030
- IMO's Greenhouse Gas Reduction Strategy and NORDEN's climate goal in alignment with the Paris Agreement of reducing transport work CO₂ emission reduction of 40% by 2030 compared to 2008 and a total CO₂ emission reduction of 50% in 2050

Environment

Shipping carries risks to the environment through spills, discharges and emissions to air, land and water. Managing these outputs is critical to environmental protection, the industry, to NORDEN and to our customers. Monitoring and management tools are key in minimising the environmental impact of NORDEN's activities and ensuring compliance with local and international regulations.

NORDEN's SO_x and NO_x emissions increased with 9 and 7%, respectively, compared to 2018. The SO_x increase is due to a higher sulphur average in the fuel consumed combined with increased activity, measured in transport work. Transport work increased by 7% compared to 2018.

Relative reduction of CO₂

Since the IMO has yet to define the calculation method for relative emissions, to represent the relative emissions, NORDEN uses Energy Efficiency Operational Indicator (EEOI), a measure of efficiency expressed in level of relative emissions to cargo transported.

EEOI is easily affected by change in vessel type composition, fuel type and speed. Vessel size determines the amount of cargo that can be transported. Even though smaller vessels might be the right mode of transport, they are less energy-efficient compared to bigger vessels.

Activity increase is also the reason for the higher NO_x emissions which is directly proportional to the increase in CO₂ emissions.

Sulphur IMO 2020

In order to reduce sulphur emissions on a global scale, IMO decided on a maximum emission of 0.5% sulphur limit on vessels, as opposed to 3.5% before. These sulphur emission levels have been in effect since 1 January 2020.

NORDEN's strategy to comply with the regulation is to switch to new fuel on some vessels and installing scrubbers on others, including installing 20 scrubbers on our owned vessels and 16 on long-term chartered vessels.

NORDEN's vessel type composition changes continuously. Therefore, we have split up the data points by vessel types in dry cargo. In tankers, the vessel type composition does not vary greatly, hence, we show the consolidated progress for tankers.

The percentages show how far NORDEN has come in reaching our 2030 target, and assuming a linear reduction, how far in reaching today's target.

- Handysize: 34% and 69%
- Panamax: 26% and 52%
- Supramax: 17% and 33%
- Tankers: 14% and 28%

Plastic

During 2019, Plastic Change visited NORDEN's offices in Hellerup to discuss the issues of plastics in general and plastics in the ocean. After the visit, NORDEN completed an internal plastic policy including a number of action points to address waste handling and reduction of use of plastics in the offices. The purpose of the policy is to create awareness internally addressing waste and plastics in a more responsible manner.

Environment

NORDEN's ambition

- 0 oil spills into water
- Ensure continuous sulphur emission compliance

Performance 2019

Emissions

- SO_x: 54,400 tonnes, an increase of 9% from 2018
- NO_x: 128,500 tonnes, an increase of 7% from 2018

Looking ahead

- Continue 0 oil spills into water
- Continue to investigate the possibility of NORDEN vessels' contribution to plastic pollution data collection

Relevant policies

- Safety & Environmental Protection Policy: All employees in every function execute their work under safe and healthy conditions and with proper concern for protecting the environment. NORDEN aims to eliminate possible dangers which may result in personal injury, professional illness, accidents, damage to property and damage to the environment
- NORDEN's Plastic Policy sets out principles for how we work with waste reduction through various initiatives phasing out products that cause unnecessary plastic waste and other waste.

Sustainability risks

NORDEN's value chain is global and complex. It comprises dry and tanker shipping, ports and yards, etc. This is a description of material sustainability issues in NORDEN's value chain.

Value chain	Risk	Mitigation measure
 Port operations	NORDEN could be facilitating the occurrence of significant negative impacts related to the transported cargo or the actors that produce or purchase the cargo. During port calls, NORDEN receives provisions, spare parts and bunker fuel from suppliers, if needed. There is a risk that NORDEN engages with suppliers who fail to live up to NORDEN's standards. Ship agents typically arrange for all support activities in connection with loading and discharge operations, including coordination with local authorities on NORDEN's behalf. This makes the Group vulnerable to sub-standard performance of agents when they represent NORDEN, and there is a risk of engaging with counter parties who have other standards than NORDEN's.	NORDEN's Anti-Corruption Compliance Programme and daily screening of NORDEN third-party commercial counter-parts on various potential risk factor issues, including sanctions lists, global law enforcement lists, etc.
 Shipping	Vessels are high-risk workplaces, and severe injuries can happen if instructions are not followed at all times. Seaborne transport involves safety and potentially also security risks. A risk assessment determines which mitigation measures to utilise prior to passage in high-risk areas. Discharges from vessels, such as oil, waste, ballast water and emissions could not be managed in line with local and global standards, potentially impacting ecosystems negatively and posing a risk to human health. External charterer could not adhere to NORDEN standards when operating NORDEN's vessels. In the long term, climate change will influence NORDEN's operations. Increased water levels result in a higher transported mass as well as new routes, but climate change also causes an increased frequency of hurricanes, flooded ports and delays. Production costs may increase, thereby changing the demand for shipping of certain products.	Anti-corruption & Ethics Training, Safety & Environmental Protection Policy, Human Rights Policy. Long-term climate change risk guidance to the Board of Directors and Senior Management on an annual basis as well as planned agenda item on Board strategy meeting in June 2020. Climate target is aligned with the UN IMO.

Value chain	Risk	Mitigation measure
 Shipping (continued)	A ship that has responded to a call of distress and rescued a group of individuals can encounter challenges with health and safety during boarding of the rescue mission as well as on board caused by limited capacity. Situations including people in distress at sea are responded in accordance with SOLAS and IMO guidelines.	Safety & Environmental Protection Policy and Best Management Practices to Deter Piracy v5 standard, including how to use anti-piracy equipment.
 NORDEN's supply chain	The levels of anti-corruption, environment and health and safety management systems are below NORDEN's standards.	Under IMPA ACT, NORDEN initiates the screening of 5 tier-1 suppliers per year.
 Sale, redelivery of chartered tonnage and recycling	Vessels are redelivered or sold to other shipowners for further trade before recycling becomes relevant. In the sale or redelivery of chartered vessels, shipowners may choose to dispose of the vessel in an unsustainable way shortly after sale.	Responsible Recycling Policy.
 Second-hand tonnage	Vessels bought by NORDEN from other shipowners could fail to live up to NORDEN's quality standards on vessel performance and environmental footprint.	Quality pre-assessment of the vessels for acquisition.
 Chartering	Compliance with NORDEN's integrity standards may not be safeguarded in commercial contracts and activities, including scenarios where NORDEN may engage with business partners who fail to live up to NORDEN's standards when taking on NORDEN's business around the world.	Anti-corruption & Ethics Training, Safety & Environmental Protection Policy.
 Newbuildings	Safety and adequate labour rights adherence at the yards and potential negative impacts on the environment from yard activities.	Risk assessment and inspections of yards.

ESG performance 2019

	2019	2018	2017
ENVIRONMENTAL PERFORMANCE			
Energy Efficiency (gCO₂/tonnes-mile)			
EEOI	8.70	8.60	8.75
Energy consumption			
Heavy fuel oils (1,000 tonnes)*	1,134	1,083	1,086
Distillate fuel oils (1,000 tonnes)*	156	142	117
Very low sulphur residuals (1,000 tonnes)	17	-	-
Electricity (MWh)	1,233	1,276	1,283
Total energy consumption (TJ)*	52,988	49,611	48,652
Transport work (1,000,000 tonnes cargo x mile)			
Total operated vessels*	469,897	445,758	429,864
CO₂ emissions (1,000 tonnes)			
Direct CO ₂ emissions (Scope 1 GHG protocol)*	4,088	3,829	3,757
Indirect CO ₂ emissions (Scope 2 GHG protocol)**	0.2	0.4	0.4
Other indirect CO ₂ emissions (Scope 3 GHG protocol)*	3.6	3.0	3.3
Total CO ₂ emissions*	4,092	3,832	3,761
Other emissions (1,000 tonnes)			
SOx emissions	54	50	53
NOx emissions*	129	120	118
Spills			
Spills > 1 barrel	0	0	0

	2019	2018	2017
SOCIAL PERFORMANCE			
Employees			
Employees at sea (FTE)*	715	878	874
Employees on shore (FTE)	395	361	346
Nationalities represented (of total workforce)	32	30	26
Women in management on shore (%)	26	22	24
Gender split at sea			
– gender with the lowest representation ♀ (%)	1	1	-
Gender split on shore			
– gender with the lowest representation ♀ (%)	35	35	33
Safety			
Rest hour non-conformity (hours per FTE)	0.25	0.33	0.34
Fatalities	0	0	1
Lost time injury frequency (million working hours)	1.45	0.29	1.58
GOVERNANCE PERFORMANCE			
Shareholder-elected board members			
Gender with the lowest representation ♀ (%)	33	33	17
Attendance (%)***	97	94	-

* New accounting policy since 2019, data is not comparable to previous years

** Updated MWh to CO₂ factor used in 2019

*** New accounting policy since 2019. In 2018, phone and strategy meetings are out of scope

Covered by PwC's 2019 report on non-financial data

ESG accounting policies

Boundary setting

The CSR report boundary includes assets and employees in the parent company and in subsidiaries as well as assets in joint ventures (JVs). Assets in JVs include vessels owned by NORDEN together with partners in the company NORD SUMMIT Pte. Ltd. Common to the vessels in JVs is that they are operated by NORDEN, but all material decisions regarding the commercial, operational and technical management of the vessels are made in agreement with the partners.

The reporting boundary

- Owned vessels
- Operated vessels
- Employees on shore
- Employees at sea
- NORDEN's site offices

Data quality and data collection

Collecting information and data on NORDEN's sustainability and ESG performance, the reporting principles of balance, clarity, accuracy, reliability, timeliness and comparability are applied. Data includes comparable data from the past 3 years; 5 if part of key figures. Significant variations in data are explained in the relevant section of the sustainability report. The sustainability assurance provider, PwC, provides independent limited assurance on NORDEN's ESG performance indicators. The ESG report includes performance data from the CSR report supplemented by additional indicators and accounting policies.

Changes to reported ESG data compared with 2018

Employees at sea (FTE) in 2019 is aligned with the financial scope where seafarers on internal and external contracts are included, and seafarers between contracts are excluded. For calculating retention and turnover, the scope for employees at sea (FTE) is defined as seafarers on internal vessels, including seafarers between contracts.

Due to new data management, accounting for bunker fuel consumption for 2019 has changed slightly from 2018 and prior. Data from 2019 and 2018 is therefore not directly comparable.

With our new asset-light business model, we find that reporting on consolidated transport work is more relevant than previously where we distinguished between owned and operated vessels. Further, it has become more relevant to report on efficiency which is why we include a consolidated EEOI in performance data overview.

In the 2019 ESG performance overview, we have also added governance indicators, including board-level gender diversity and attendance.

Environmental performance

NORDEN is liable for consumption, emissions and other environmental elements. For owned assets which are leased out, NORDEN is not liable for consumption, emissions and other environmental elements – the lessee is.

Energy consumption

Transport work (1,000,000 tonnes cargo x mile): Transport work of owned and operated vessels as well as vessels chartered and operated. Transport work expresses the mass of cargo transported over distance, both as registered in the Integrated Maritime Operations System IMOS, NORDEN's commercial maritime chartering, operations and financial solution. Cargo transported over distance, compared with the amount of CO₂ emitted, provides a picture of fleet efficiency – see EEOI.

Heavy fuel oil and distillate fuel oil and very low sulphur residuals (1,000 tonnes): Bunker fuel consumed directly by owned and operated vessels, including all the fuels, gas oils and residuals used on board for activities such as steam, generating electricity,

marine boilers, heating, main and auxiliary engines. In Norient Product Pool, consumed fuel accounted for includes the vessels controlled by NORDEN. Total daily consumption reported by vessels into IMOS validated with bunker acquisitions and deviations is regularly controlled.

Electricity (MWh): Electricity and heat consumed indirectly in operational activities for onshore offices. Based on annual consumption in Brazil, China, Denmark, India, Ivory Coast, Singapore offices. Excludes Australia, Canada, Chile, Cyprus and USA offices as utilities are included in the rent.

Total energy consumption (TJ): calculated adding up tonnes fuel and electricity use, using heating value 40.2MJ/kg for heavy fuel oil, 42.7MJ/kg for distillate fuel oil, 41.7MJ/kg for very low sulphur residuals and 0.0036 MWh for electricity.

Energy Efficiency Operational Indicator (gCO₂/cargo-mile): The Energy Efficiency Operational Indicator (EEOI) is a measurement of efficiency and is defined as the amount of CO₂ emitted per tonne of cargo transported 1 mile based on fuel consumption and transport work.

CO₂ emissions (1,000 tonnes): Compatible with the Greenhouse Gas (GHG) Protocol

- Scope 1: Direct emissions from NORDEN's own consumption of fuel from owned company cars and energy from owned and chartered vessels
 - o Chartered and owned vessels: Based on fuel consumption for the year
 - o Owned company cars: per 31 December 2019 based on conversion Key2Green conversion factor 2.65 kg CO₂/L
- Scope 2: Indirect emissions from purchased electricity and district heating as well as for owned electrical car assuming 15kWh/100km.

- Scope 3: Indirect emissions from third party activities. Measures CO₂ emissions from air travel and leased company cars.
 - o Air travel: Route < km 1,000 = CO₂ 0.18 per km and routes > km 1,000 = CO₂ 0.11 per km
 - o Leased cars: per 31 December 2019 based on conversion Key2Green conversion factor 2.65 kg CO₂/L

NOx and SOx emissions (1,000 tonnes)

- NOx: Nitrogen oxide emissions from combustion of fuels from owned and operated vessels. NO₂ emissions from the energy produced by main engine x Tier I NOx limit (i.e. 17 g/kWh).
- SOx: Sulphur oxide emissions mainly stem from the burning of the sulphur compound in the fuel from owned and operated vessels. SO₂ emissions are calculated from the fuel quantity consumed during the year multiplied by the average sulphur content in the bunker fuel purchased by Bunkers Department.

Spills (> 1 barrel): includes incidents of oil spills larger than 1 barrel (159 litres) into the sea from owned and/or operated vessels. Any oil spills are registered in NORDEN's fleet and vessel performance system K-Fleet/C-experience.

Social performance

Scope and boundaries

Scope for FTEs at sea includes areas where NORDEN has the direct governing authority and responsibility for safety and environmental management of people, processes, facilities and vessels. Out of scope: Non-owned vessels and vessels on bare-boat charter.

Scope for full-time workforce FTEs on shore includes permanent and time-limited employees (fix-term, student job and temporary) in NORDEN's offices, except for indicators Retention and Turnover where scope in-

cludes average FTE on shore of permanent employees and excludes fix-term, student job and temporary.

Employees

Employees at sea (FTE): Average full-time equivalent number of employees at sea during the year. Scope for 2017-2018 includes employees on NORDEN's owned vessels, including average seafarers employed on permanent and fixed-term contracts as defined in NORDEN's crew management system. Scope for 2019 includes seafarers on internal and external, excl. seafarers between contracts. Note scope 2017-2018 is used to calculate retention and turnover in the People section.

Employees on shore (FTE): Average full-time equivalent number of employees on shore as defined in NORDEN's HR system (Fairsail). Scope for employees, women in management, gender split: Average FTE calculation includes permanent and time-limited employees (fix-term, student job and temporary).

Nationalities represented (of total workforce): Number of nationalities in total workforce on shore (based on NORDEN's HR system Fairsail) and at sea for owned vessels in internal management (based on crew management system) during the accounting year.

Women in management on shore (%): Average FTE number of women in management positions on shore out of the total pool of average FTE onshore managers. Managers are defined as a person with responsibility for a team of at least 1 other employee, subject to NORDEN's employee development programme (excluding students, interns, etc.) as defined in NORDEN's HR system (Fairsail).

Gender split at sea, gender with the lowest representation ♀ (%): The percentage of the average number of women FTE at sea out of a total average number of FTEs at sea during the year based on crew management system.

Gender split on shore, gender with the lowest representation ♀ (%): The percentage of the average number of women FTE on shore out of a total average number of FTEs at sea during the year based on NORDEN's HR Management System Fairsail.

Health and safety

Rest hour non-conformity (RHNC) at sea (hours per FTE): Rest hour non-conformities is defined as number of days where a seafarer is not in compliance. The number of non-compliance days per 30-working day period on board. Not more than 1 violation is registered per unique person in a 24 hour period. Rest hour non-conformity is monitored onboard in NORDEN's crew management system in accordance with the ILO and STCW conventions. Scope for RHNC includes FTEs on owned operated vessels in own management.

Fatalities: Number of fatalities in the reporting year registered in NORDEN's technical administration system.

Lost Time Injury Frequency at sea (injuries per million working hours): Calculated based on the number of work-related accidents registered in NORDEN's technical management system, which causes a seafarer to be unable to work for more than 24 hours (based on exposure hours), in accordance with OCIMF's 'Marine Injury Reporting Guidelines', per one million exposure

hours. Scope for LTIF includes FTEs on owned vessels in own or external management.

Governance

Shareholder-elected board members: Gender with the lowest representation ♀ (%): Percentage of shareholder-elected women in the Board of Directors out of total number of shareholder-elected Board members at year-end.

Attendance (%): Attendance rate at Board meetings (including phone and strategy meetings) attended by shareholder-elected board members throughout the year. Scope is excluding employee-elected board members. Scope for 2018 excludes phone and strategy meetings